Translational AI Center

# AI Integration Workshop

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# Hello, I'm Aditya

**Data Scientist · Translational AI Center, Iowa State University**

## Research focus

- **Physics-informed ML —** domain knowledge + data
- **GPU computing —** scalable scientific computation
- **Neural PDE solvers —** data-driven physics
- **Decentralized learning —** communication-efficient ML
- **Computer-aided engineering —** deep learning for design

## Background

- **PhD —** ME + Computer Engineering, ISU (2020)
- **BE (Hons) —** Mechanical Engineering, BITS Pilani
- **Industry —** Ansys ML co-op; design engineer at FMC
- **Impact —** 1,500+ citations; h-index 18
- **Service —** Excellence in Service awardee, TrAC 2026

# Why I'm here today

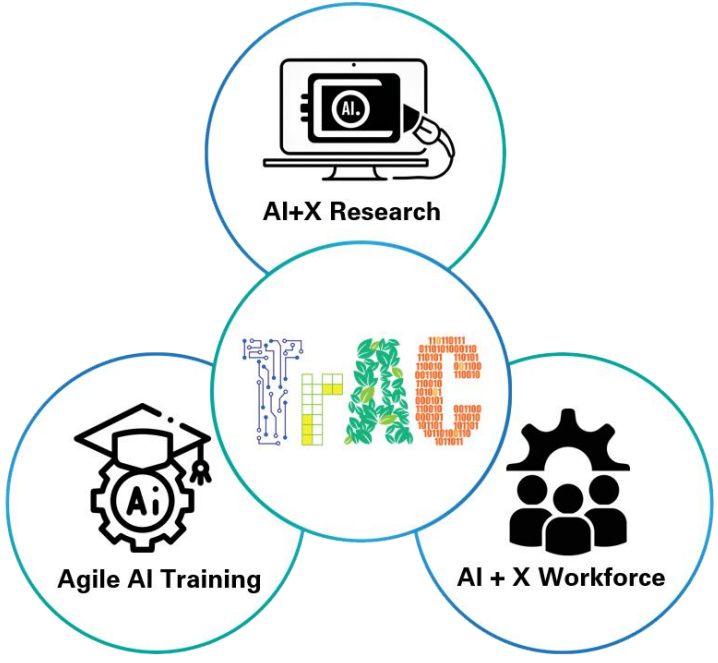
From research labs to your boardroom — translating AI into business strategy.

- **Bridge the language gap —** we'll demystify AI without dumbing it down
- **Frameworks over hype —** you'll leave with tools, not just talking points
- **Industry-tested —** examples grounded in manufacturing, agriculture, design, education
- **Active engagement —** expect quizzes, exercises, and table discussions
- **Take-home assets —** AI Integration Toolkit, scorecard, opportunity matrix template

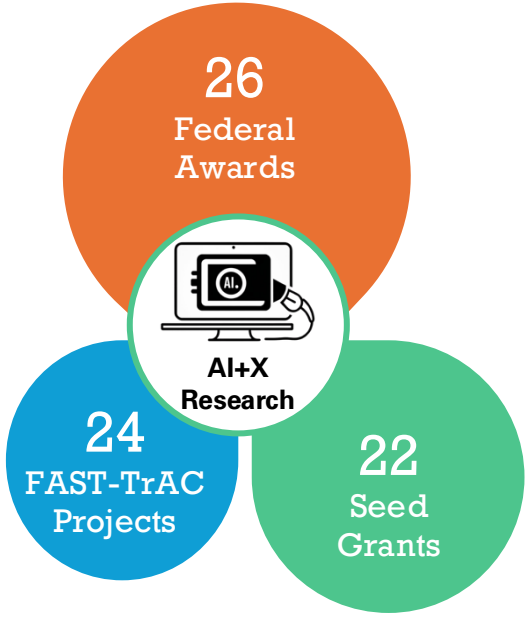
# Translational AI Center

- Iowa State's AI research hub
- A pre-compete industrial nexus for Translational AI
- 70+ affiliated faculty members across 7 ISU colleges

# TrAC Objectives



# TrAC AI+X Research




DARPA BIO-SHIELD

AI-AgARENA
*AI-ready Cyber-Ag System Testbed*

**AI-ENGAGE**

SciML-Accelerated Cardiac Digital Twin

# Hands-on AI Consulting Service

FAST-TrAC

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# Introduction to AI

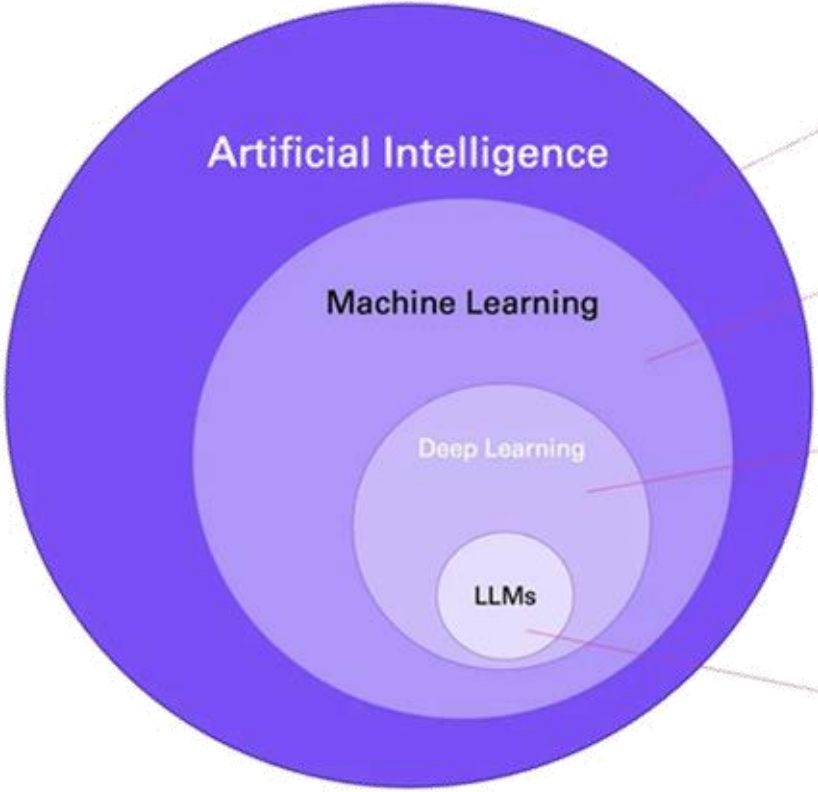
The vocabulary, history, and shape of the field.

# AI in a gist

Teacher Children

Data Scientist AI

# Definitions



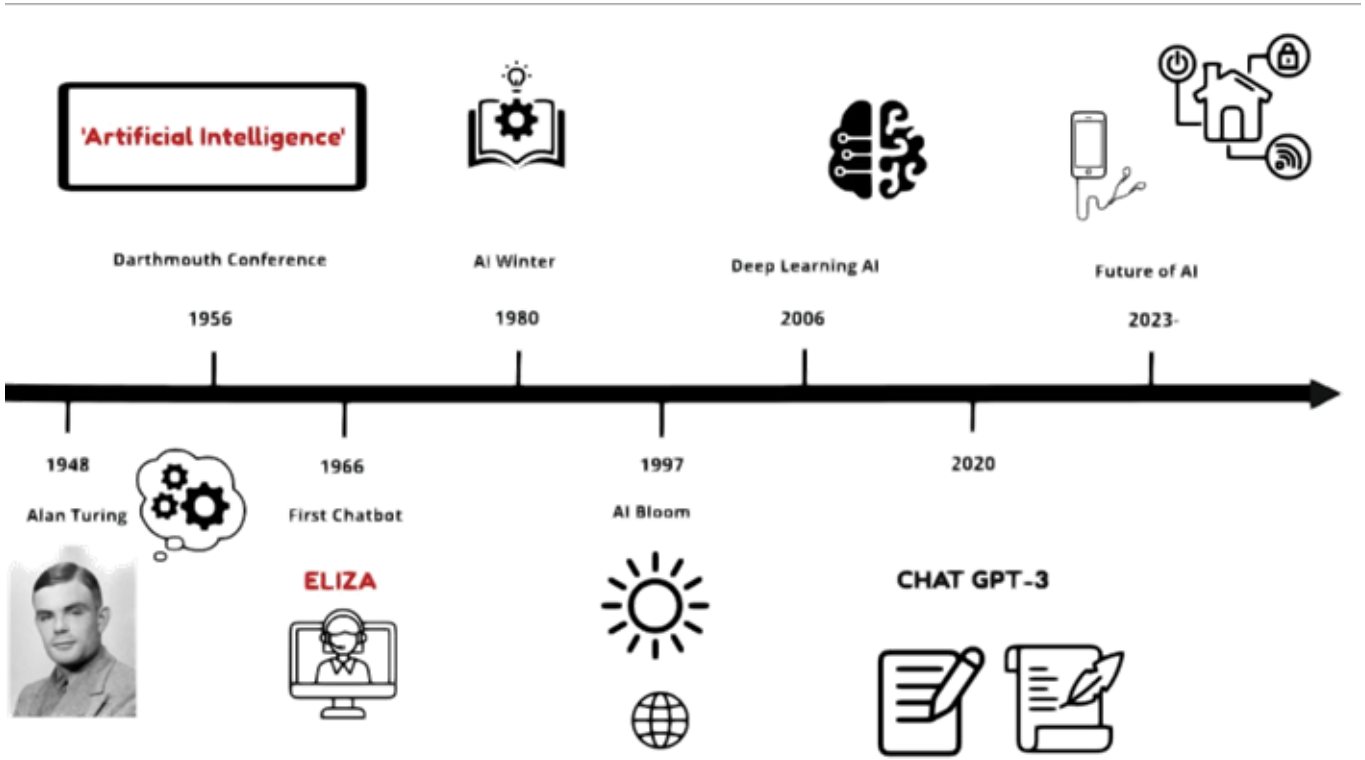

**Intelligent Machines**
Broadly defined

**Pattern Recognition**
Learning general patterns from data

**Neural Networks**
Learning general patterns in **unstructured** data (i.e. images, text, audio, etc.)

**Large Language Models**
Learning to understand natural language (i.e. text)

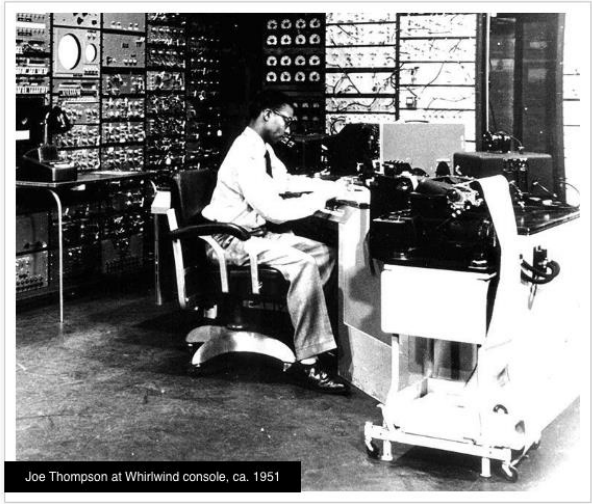
# History of AI



# Constrasting with History of Computers

## Direct keyboard input to computers

1956

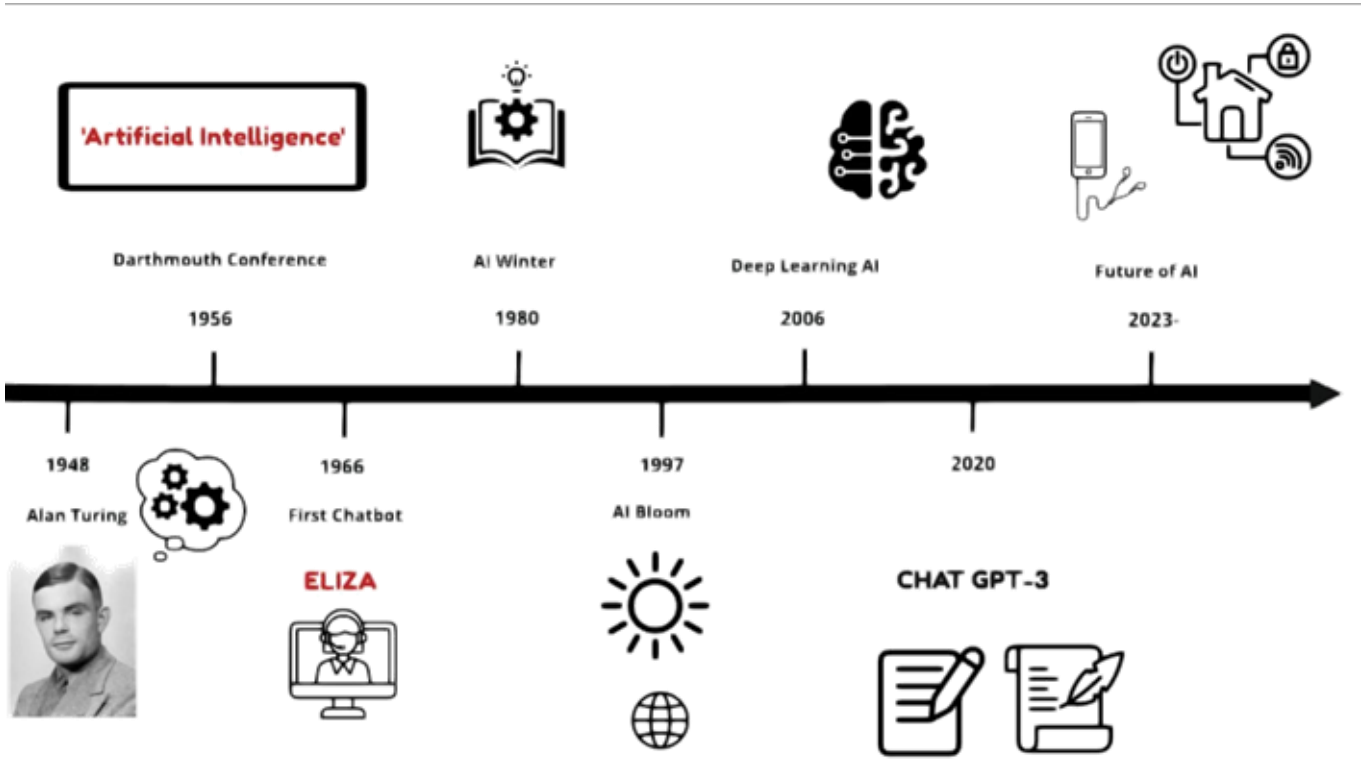


Joe Thompson at Whirlwind console, ca. 1951

1957

At MIT, researchers begin experimenting with direct keyboard input to computers, a precursor to today´s normal mode of operation. Typically, computer users of the time fed their programs into a computer using punched cards or paper tape. Doug Ross wrote a memo advocating direct access in February. Ross contended that a Flexowriter -- an electrically-controlled typewriter -- connected to an MIT computer could function as a

# History of AI



# The Evolution of AI Systems

| Mental Model | | Examples |
|---|---|---|
| • Still prediction<br>• Applied to creation | **Generative AI** | • Text<br>• Image<br>• Audio<br>• Video |
| • Still prediction-based<br>• Focused on decision support | **Predictive AI** | • Forecasting<br>• Recommenders<br>• Classification |
| • Not explicitly programmed<br>• Learns patterns from data | **Learning Systems** | • Machine Learning<br>• Neural Networks |
| • Still deterministic<br>• Appears intelligent<br>• Limited flexibility | **Deterministic AI** | • Expert Systems<br>• Encoded Logic |
| • Fully programmed<br>• No learning<br>• Predictable behavior | **Deterministic Systems** | • Standard Programs<br>• Consistent Logic |

# Generative vs Predictive AI

## Generative AI

Creates new content from patterns

- **Asks** — what does X look like?
- **Output** — text, image, audio, video, code
- **Strength** — creativity, drafting, exploration
- **Watch out** — hallucinations, IP, factuality
- **Tools** — GPT, Claude, Gemini, Stable Diffusion

## Predictive AI

Forecasts outcomes from patterns

- **Asks** — what will happen / which class?
- **Output** — score, label, probability, forecast
- **Strength** — decision support, automation
- **Watch out** — data drift, training data bias
- **Tools** — XGBoost, regression, classifiers, NN

# Deterministic vs Learning Systems

Where does AI start, and where does "just code" end?

## Deterministic

- **Behavior** — rules, formulas, hand-coded logic
- **Predictable** — same input → same output, every time
- **Examples** — spreadsheets, ERP rules, expert systems
- **Hard part** — humans must enumerate every case
- **Limit** — doesn't scale to messy real-world inputs

## Learning

- **Behavior** — patterns inferred from data
- **Probabilistic** — same input may produce slight variations
- **Examples** — image classifiers, LLMs, recommender systems
- **Hard part** — data quality, drift, bias
- **Limit** — needs supervision and monitoring

# 3 Ingredients of AI

Algorithmic Advancements

Explosion of Data

Computing power

Modern AI

# Modalities of Data & Tasks

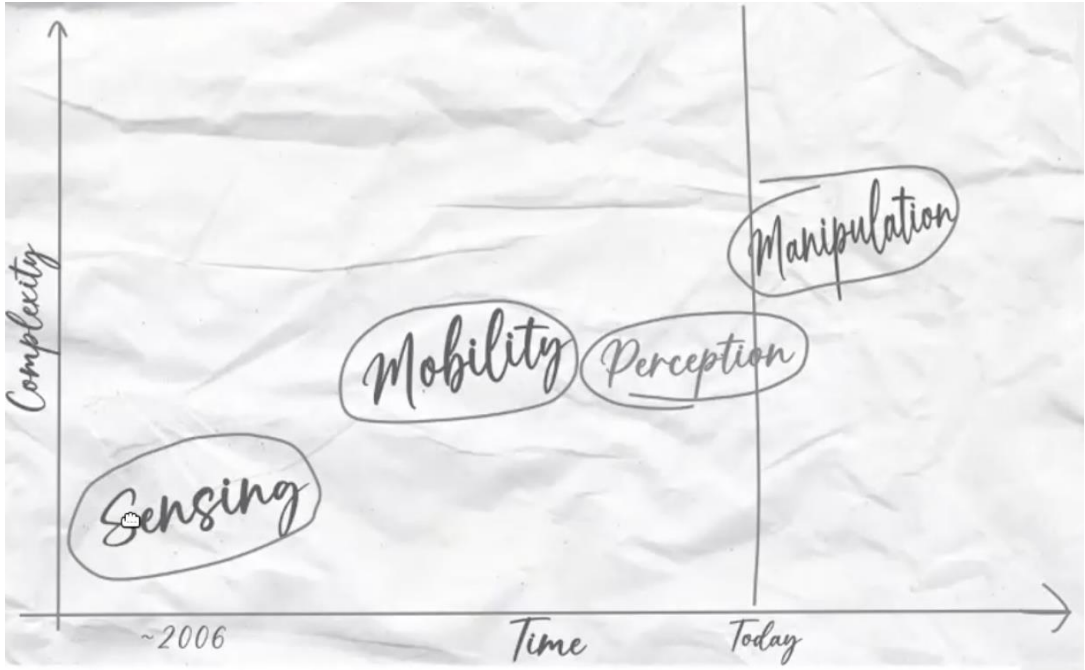
Images/Videos (or Computer Vision)

3D Vision

Speech/Audio

Text – natural language

Sensor Data



# Generative Pretrained Transformer - GPT

- 2018 OpenAI releases a paper outlining ‘GP’ or generative pretraining.
- 2019 OpenAI releases GPT-2 a 1.5 billion parameter model.
- 2020 OpenAI releases GPT-3 a 175 billion parameter model.
- 2022 OpenAI releases ChatGPT a GPT-3 model fine-tuned with Reinforcement Learning from Human Feedback.

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# From Personal Tool to Company Strategy

Why your AI as a productivity hack ≠ your company's AI strategy.

# The Fundamental Mindset Shift

Most leaders enter AI thinking like users. Strategy demands thinking like operators.

## AI as a Personal Tool

- **Goal** — save my time
- **Scope** — my workflow, my tasks
- **Skill** — good prompting
- **Risk** — low — only my output is affected
- **Win** — I'm 30% faster

## AI as a Company Strategy

- **Goal** — create durable advantage
- **Scope** — products, processes, people, partners
- **Skill** — use-case selection + change management
- **Risk** — real — affects customers, regulators, brand
- **Win** — the company is structurally better at its mission

# Operational vs Strategic AI

Both matter. Most failures come from confusing one for the other.

## Operational AI

Excellence Through Efficiency

- **Question** — how do we do today's work better?
- **Outcome** — lower cost, higher throughput, fewer errors
- **Examples** — predictive maintenance, doc extraction, defect detection
- **Owner** — operations leader; ROI-driven
- **Risk** — narrow scope, modest upside

## Strategic AI

Transformation Through Innovation

- **Question** — what new can we do that we couldn't before?
- **Outcome** — new products, new markets, new business models
- **Examples** — generative design, autonomous services, personalized offerings
- **Owner** — executive team; strategy-driven
- **Risk** — longer payback, real failure rate

# Self-Assessment: Activate Your Foundation

Before building new frameworks, let's activate what you already know. This diagnostic quiz serves three critical purposes:

### Refresh Core Concepts

Reactivate Fluency principles we'll reference throughout today—prompt engineering, context techniques, understanding AI capabilities and limitations.

### Establish Your Baseline

Identify where you stand in AI knowledge. This isn't evaluative—it's diagnostic, revealing which concepts need reinforcement.

### Prime Strategic Thinking

Retrieval practice strengthens learning and prepares you to apply knowledge in new strategic contexts.

The quiz covers five domains: prompt engineering fundamentals, context and specificity, AI tool selection, understanding limitations, and recognizing appropriate use cases. Score 80%+ across domains? Excellent foundation. Lower scores? We'll reinforce those areas today.

What is the MOST important element when crafting an effective AI prompt for business applications?

A) Using polite language and saying "please" and "thank you"
B) Providing specific context and clear success criteria
C) Writing the longest, most detailed prompt possible
D) Using technical jargon to sound professional

You need AI to analyze quarterly production data. Which context provision strategy is MOST effective?

A) "Analyze our data and find insights"
B) "You are a manufacturing analyst. Analyze Q3 production data focusing on defect rates, comparing to Q2 baseline of 2.3%"
C) Upload all company data and let AI figure out what's relevant
D) Provide only the numbers without any background

Your organization wants AI-powered visual quality inspection. What factors should drive your build/buy/partner decision?

A) Always build in-house for maximum control
B) Always buy off-the-shelf for fastest implementation
C) Choose whatever IT prefers
D) Evaluate based on strategic importance, existing capabilities, cost, and time-to-value

When is AI most likely to "hallucinate" (generate plausible but false information)?

A) When it gets confused by complex questions
B) When asked about specific facts, numbers, or recent events not in its training data
C) Randomly, with no predictable pattern
D) Only when the internet connection is poor

Your team needs to extract data from 500 PDF equipment manuals. Which approach is most appropriate?

A) Have an intern copy-paste everything into ChatGPT
B) Use ChatGPT's web interface for each document
C) Implement an API-based solution with batch processing
D) This task is impossible with current AI

# Quiz Debrief — What Your Answers Reveal

There aren't right answers — there are mindsets.

- **Q1 (effective prompt) —** B — context + success criteria. Not jargon, not length.
- **Q2 (production data) —** B — provide structure, baselines, and the decision you'll make.
- **Q3 (visual inspection) —** consider lighting, edge-case classes, false-positive cost, retraining cadence.
- **Q4 (hallucinations) —** B — when the model is uncertain and isn't grounded in your data.
- **Q5 (500 PDFs) —** RAG with extraction templates beats one-off prompting.
- **Pattern —** Strategic answers usually trade speed for ground-truth quality.

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# The AI Lifecycle & Who Does What

Roles, responsibilities, and your place on the team.

# The AI Lifecycle

From business problem to value — and back again.

Define use case + success metric

Collect, clean, label, govern

Train, validate, evaluate

Integrate, scale, secure

Drift, fairness, value

Each stage feeds the next — and monitoring loops back to refine the model and reframe the problem.

# Who Does What in AI

Three role families bring an AI initiative to life.

## Technical

- Data Engineer
- Data Scientist
- ML Engineer
- MLOps / Platform
- AI Researcher

## Product & Domain

- Product Manager (AI)
- Domain SME
- UX / Service Designer
- Ethicist / Responsible AI
- Legal / Privacy / Compliance

## Leadership & Governance

- Executive Sponsor
- AI Champion (you)
- Governance Committee
- Vendor Management

# Who Should You Be in AI?

Most of you in this room are AI Champions — you don't build the models, but adoption depends on you.

1

## Champion

Drive adoption inside the organization. Translate value, manage change, build coalitions.

2

## Sponsor

Allocate budget and political capital. Remove organizational blockers.

3

## Decision-maker

Choose what to build, buy, or skip. Set the bar for risk and quality.

4

## Advocate

Bring AI to your team's daily work. Surface opportunities and friction.

# RACI Snapshot for an AI Initiative

Responsible · Accountable · Consulted · Informed

| Stage | R | A | C | I |
|---|---|---|---|---|
| **Define use case** | PM + SME | Sponsor | Engineering | Governance |
| **Build & train** | ML + Data Eng | Eng Lead | SME | PM |
| **Validate** | SME + ML Eng | PM | Ethicist | Sponsor |
| **Deploy** | MLOps + IT | PM | Legal | Sponsor + Gov |
| **Operate** | MLOps | PM | Engineering | Sponsor |

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# Are You Ready?

Honest assessment across data, infrastructure, culture, and skills.

# The AI Maturity Model

Five stages — most companies start in Stage 1 or 2.

**01**

### Aware

Leaders know AI matters but no concrete pilots

**02**

### Exploring

Pilots underway in 1-2 functions

**03**

### Adopting

Multiple production AI systems; ad-hoc governance

**04**

### Embedded

AI is core to operating model; centralized platform

**05**

### Strategic

AI shapes business strategy and differentiation

# Four Dimensions of AI Readiness

## Data

Quality, accessibility, governance, lineage

## Infrastructure

Compute, storage, MLOps, observability

## Culture

Risk appetite, learning, cross-functional collab

## Skills

Technical depth, domain SMEs, AI literacy

# EXERCISE — Readiness Scorecard

20 minutes individual; refer to the AI Integration Toolkit (Excel) on your table.

- **For each of the 4 dimensions —** rate your organization 1 (low) → 5 (high)
- **Identify your weakest —** this is your biggest investment area for the next 12 months
- **Identify your strongest —** this is your foundation to build pilots on
- **Total your score —** 0–25%: foundation sprint · 25–50%: pilot mode · 50–75%: scale · 75–100%: strategic
- **We'll debrief at the table —** in 5 minutes

# Your Strategic Foundation

By the end of Module 1 you should be able to:

- **Place AI —** in the spectrum from deterministic systems to generative AI
- **Translate —** between business goals and AI capabilities
- **Distinguish —** personal-productivity AI from strategic AI
- **Identify —** the roles needed to execute an AI initiative
- **Score —** candidate opportunities on value × effort
- **Assess —** your organization across the four readiness dimensions